

BOOK	CHAPTER	FORMULA		VARIABLES
Book 1 Foundation of Risk Management	Modern Portfolio Theory (MPT) and the Capital Asset Pricing Model (CAPM)	CAPM formula (expected return on asset i)	$E(R_i) = R_f + \beta_i(R_m - R_f)$	$\beta_i = \rho_{(im)} \frac{\sigma_i}{\sigma_{m_i}} = \frac{Cov(i, m)}{\sigma_m^2} = \frac{\sigma_{im}}{\sigma_m^2}$ R_m = expected market rate of return R_f = risk-free rate $E(R_i)$ = expected return on asset i $\rho_{im} = \frac{Cov(i, m)}{\sigma_i \sigma_m} = \frac{\sigma_{im}}{\sigma_i \sigma_m}$
		Capital market line	$E(R_p) = R_f + \frac{\sigma_p}{\sigma_m}(R_m - R_f)$	R_m = expected market rate of return R_f = risk-free rate $E(R_p)$ = portfolio expected return σ_p = portfolio standard deviation σ_m = market standard deviation
		Capital allocation line	$E(R_p) = R_f + \frac{\sigma_p}{\sigma_i}(R_i - R_f)$	R_i = expected rate of return on asset i R_f = risk-free rate $E(R_p)$ = portfolio expected return σ_p = portfolio standard deviation σ_i = standard deviation of asset

		Sharpe ratio	Sharpe ratio = $\frac{E(R_p) - R_f}{\sigma(R_p)}$	$\sigma(R_p)$ = portfolio standard deviation $E(R_p)$ = portfolio expected return R_f = risk-free rate
		Treynor ratio	Treynor ratio = $\frac{E(R_p) - R_f}{\beta_P}$	β_P = portfolio beta $E(R_p)$ = portfolio expected return R_f = risk-free rate
		The tracking error (TE)	$TE = (R_P - R_{Benchmark})$	R_P = return on the portfolio $R_{Benchmark}$ = return on the benchmark portfolio
		Sortino ratio (SR)	$SR = \frac{R_p - T}{\frac{1}{N} \sum_{t=1}^N \min(0, R_{pt} - T)^2}$	T = target or required rate of return $\frac{1}{N} \sum_{t=1}^N \min(0, R_{pt} - T)^2$ = downside deviation, as measured by the standard deviation of negative returns
		Information ratio (IR)	$IR = \frac{E(R_P - R_B)}{\sqrt{Var(R_P - R_B)}}$	R_P = return on the portfolio $R_{Benchmark}$ = return on the benchmark portfolio
	The Arbitrage Pricing Theory and Multifactor Models of Risk and Return	Return on a security	$R_i = E(R_i) + \beta_{i1}[I_1 - E(I_1)] + \dots + \beta_{iK}[I_k - E(I_k)] + e_i$	R_i = rate of return on security i $I_1 - E(I_1)$ = difference between observed and expected values in factor k β_{ik} = coefficient measuring the effect of changes in a factor I_k on the rate of return of security i e_i = noise factor (i.e., the idiosyncratic factor).

		Variance/Covariance for a factor model with M factors	$M + \frac{M^2 - M}{M}$	M = number of factors in the model
		Number of covariances required	$\frac{n^2 - n}{2}$	n = number of variances
		The Fama-French Model (FFM)	$E(R_i) = R_f + \beta_{i,MKT}E(R_m - R_f) + \beta_{i,SMB}E(SMB) + \beta_{i,HML}E(HML)$	$E(R_i)$ = expected return on stock i R_f = risk-free interest rate SMB = size factor $\beta_{i,SMB}$ = factor-beta for the size factor HML = value factor $\beta_{i,HML}$ = factor-beta for the value factor $E(R_m - R_f)$ = CAPM market factor $\beta_{i,MKT}$ = factor-beta for the market-factor
Book 2 Quantitative Analysis	Fundamentals of Probability	Mutually exclusive events	$P(A \cap B) = P(A \text{ and } B) = 0$ $P(A \cup B) = P(A \text{ or } B) = P(A) + P(B)$	$P(A \cap B)$ = probability A intersection B $P(A \cup B)$ = probability A union B
		Conditional probability	$P(A B) = \frac{P(A \cap B)}{P(B)}$	$P(A B)$ = probability of A given B $P(A \cap B) = P(A B)P(B)$
		Independent events	$P(A \cup B) = P(A \text{ or } B) = P(A) + P(B) - P(A \cap B)$	
			$P(A \cap B) = P(A \text{ and } B) = P(A) \times P(B)$	

			Conditional Probability	$P(A B) = \frac{P(A)P(B)}{P(B)} = P(A)$	
		Bayes' Theorem		$P(A B) = \frac{P(B A)P(A)}{P(B)}$	
	Univariate Random Variables	Binomial distribution		$P(X = x) = \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x}, x = 0, 1, 2, \dots, n$	$P(X = x)$ = probability mass function of X
				$F_X(x) = \sum_{i=1}^{ x } \binom{n}{i} p^i (1-p)^{n-i}$	$F_X(x)$ = cumulative distribution function of X
				$E(X) = np$	$E(X)$ = expectation of X
				$V(X) = np(1-p)$	$V(X)$ = variance of X
		Bernoulli distribution		$P(X = x) = p^x (1-p)^{1-x}, x = 0, 1; 0 < p < 1$	$P(X = x)$ = probability mass function
				$F_X(x) = \begin{cases} 0, & y < 0 \\ 1-p, & 0 \leq y < 1 \\ 1, & y \geq 1 \end{cases}$	$F_X(x)$ = cumulative distribution function
				$E(X) = p$	$E(X)$ = expectation of X
				$V(X) = p(1-p)$	$V(X)$ = variance of X

		Uniform distribution	$P(X = x) = \frac{1}{n}, x = 1, 2, \dots, n$	$P(X = x)$ = probability mass function
			$E(X) = \frac{n+1}{2}$	$E(X)$ = expectation of X
			$V(X) = \frac{n^2 - 1}{12}$	$V(X)$ = variance of X
		Poisson distribution	$P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}, x = 0, 1, 2, \dots; \lambda > 0$	$P(X = x)$ = probability mass function
			$F_X(x) = e^{-\lambda} \sum_{i=1}^{ x } \frac{\lambda^i}{i!}$	$F_X(x)$ = cumulative distribution function of X
			$E(X) = V(X) = \lambda$	$E(X)$ = expectation of X
		Continuous uniform distribution	$f_X(x) = \frac{1}{b-a}, a \leq X \leq b$	$f(x)$ = probability density function of X
			$F_X(x) = \begin{cases} 0, & x < a \\ \frac{x-a}{b-a}, & a \leq x \leq b \\ 1, & x \geq b \end{cases}$	$F_X(x)$ = cumulative distribution function
			$E[X] = \frac{a+b}{2}$	$E(X)$ = Expectation of X
			$V[X] = \frac{(b-a)^2}{12}$	$V(X)$ = Variance of X

		Normal distribution	$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, -\infty < X < \infty$	$f(x)$ = probability density function of x
			$E(X) = \mu$	$E(X)$ = expectation of X
			$V(X) = \sigma^2$	$V(X)$ = variance of X
		Log-normal distribution	$f(x) = \frac{1}{x\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{\ln x - \mu}{\sigma}\right)^2}$	$f(x)$ = probability density function of x
			$V(X) = (e^{\sigma^2} - 1)e^{2\mu + \sigma^2}$ $E[X] = e^{\mu + \frac{1}{2}\sigma^2}$	
			$F_X(x) = \Phi\left(\frac{\ln X - \mu}{\sigma}\right)$	
		Chi-square distribution	$f(x) = \frac{1}{2^{\frac{k}{2}}\Gamma\left(\frac{k}{2}\right)} x^{\frac{k}{2}-1} e^{-\frac{x}{2}}, x > 0$	$\Gamma(n) = \int_0^{\infty} x^{n-1} e^{-x} dx$
			$E(S) = k$ $V(S) = 2k$	$S = \sum_{i=1}^k Z_i^2$

		Student's t-distribution	$f(x) = \frac{\Gamma\left(k + \frac{1}{2}\right)}{\sqrt{k\pi} \Gamma\left(\frac{k}{2}\right)} \left(1 + \frac{x^2}{k}\right)^{-\frac{k+1}{2}}$	$X = \frac{Z}{\sqrt{\frac{U}{k}}}$ Z = standard normal variable, U = a chi-square variable with k degrees of freedom
			$E(X) = 0$ $V(X) = \frac{k}{k-2}$	
		F-distribution	$f(x) = \frac{\sqrt{\frac{(k_1 X)^{k_1} k_2^{k_2}}{(k_1 X + k_2)^{k_1 + k_2}}}}{xB\left(\frac{k_1}{2}, \frac{k_2}{2}\right)}$	$X = \frac{U_1/k_1}{U_2/k_2} \sim F(k_1, k_2)$ U_i = chi-squared distributions that are independent with k_i degrees of freedom.
		Beta distribution	$f(x) = \frac{1}{B(a, b)} x^{a-1} (1-x)^{b-1}, \quad 0 \leq x \leq 1$	$B(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$
			$E(X) = \frac{a}{a+b}$	
			$V(X) = \frac{ab}{(a+b)2(a+b+1)}$	
		Exponential distribution	$f_X(x) = \frac{1}{\beta} e^{-\frac{x}{\beta}}, x \geq 0$	
			$F_X(x) = 1 - e^{-\frac{x}{\beta}}$	
			$E(X) = \beta$	
			$V(X) = \beta^2$	

	Multivariate Random Variables	PMF of a trinomial random variable	$f_{X_1, X_2}(x_1, x_2) = \frac{n!}{x_1! x_2! (n - x_1 - x_2)!} p_1^{x_1} p_2^{x_2} (1 - p_1 - p_2)^{n - x_1 - x_2}$	
		CDF of a bivariate random variable	$F_{X_1, X_2}(x_1, x_2) = \sum_{\substack{t_1 \in R(X_1) \\ t_1 \leq x_1}} \sum_{\substack{t_2 \in R(X_2) \\ t_2 \leq x_2}} f_{(X_1, X_2)}(t_1, t_2)$	
		Marginal distribution	$f_{X_1}(x_1) = \sum_{x_2 \in R(X_2)} f_{X_1, X_2}(x_1, x_2)$	$f_{X_1}(x_1)$ = marginal distribution of X_1
			$f_{X_2}(x_2) = \sum_{x_1 \in R(X_1)} f_{X_1, X_2}(x_1, x_2)$	$f_{X_2}(x_2)$ = marginal distribution of X_2
		Conditional distribution	$f_{X_1 X_2}(x_1 X_2 = x_2) = \frac{f_{X_1}(x_1) f_{X_2}(x_2)}{f_{X_2}(x_2)}$	X_1 and X_2 are independent $f_{X_1, X_2}(x_1, x_2) = f_{X_1}(x_1) f_{X_2}(x_2)$
		Expectation of ($g(X_1, X_2)$)	$E(g(X_1, X_2)) = \sum_{x_1 \in R(X_1)} \sum_{x_2 \in R(X_2)} g(x_1, x_2) f_{X_1, X_2}(x_1, x_2)$	
		Moments	1 st moment: $E(X) = [E(X_1), E(X_2)] = [\mu_1, \mu_2]$ 2 nd moment: $Var(X_1 + X_2) = Var(X_1) + Var(X_2) + 2Cov(X_1, X_2)$	$COV(X_1, X_2) = E[X_1 X_2] - E[X_1]E[X_2]$

		Correlation	$\text{Corr}(X_1, X_2) = \rho_{X_1 X_2} = \frac{\text{Cov}(X_1, X_2)}{\sqrt{\sigma_1^2} \sqrt{\sigma_2^2}} = \frac{\sigma_{12}}{\sigma_1 \sigma_2}$	$\text{Corr}(X_1, X_2)$ = correlation coefficient between X_1 and X_2 $\text{Var}(X_1) = \sigma_1^2$ $\text{Var}(X_2) = \sigma_2^2$ $\text{Cov}(X_1, X_2) = \sigma_{12}$
		The variance of a sum of two random variables	$\text{Var}(X_1 + X_2) = \text{Var}(X_1) + \text{Var}(X_2) + 2\text{Cov}(X_1, X_2)$	
			$\text{Var}(aX_1 + bX_2)$ $= a^2\text{Var}(X_1) + b^2\text{Var}(X_2)$ $+ 2ab\text{Cov}(X_1, X_2)$	a and b are constants
		The variance of a sum of two random variables if X_1 and X_2 are independent	$\text{Var}(X_1 + X_2) = \text{Var}(X_1) + \text{Var}(X_2)$	
		Conditional expectations	$E(X_1 X_2 = x_2) = \sum_i x_{1i} f(X_1 X_2 = x_2)$	$E(X_1 X_2 = x_2)$ = conditional expectation of X_1 given $X_2 = x_2$
		Conditional variance	$\text{Var}(X_1 X_2 = x_2) = E(X_1^2 X_2 = x_2) - [E(X_1 X_2 = x_2)]^2$	$\text{Var}(X_1 X_2 = x_2)$ = conditional variance of X_1 given $X_2 = x_2$
		The joint pdf	$P(a < X_1 < b, c < X_2 < d)$ $= \int_a^b \int_c^d f_{X_1, X_2}(x_1, x_2) dx_1 dx_2$ $\text{for } f_{X_1, X_2}(x_1, x_2) \geq 0$	
		Mean of (iid) random variable	$E\left(\sum_i^n X_i\right) = \sum_i^n E(X_i) = \sum_i^n \mu = n\mu$	$E(X_i) = \mu$

		Variance of (iid) random variable	$Var\left(\sum_i^n X_i\right) = \sum_i^n \sigma^2 + 2 \sum_{j=1}^n \sum_{k=j+1}^n 0 = \sum_i^n \sigma^2 = n\sigma^2$	$X_i^{iid} \sim N(\mu, \sigma^2)$
	Sample Moments	Population mean	$\mu = \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$	$\bar{X}$ = sample mean μ = the population mean
		Expectation of the mean estimator, $\hat{\mu}$	$E(\hat{\mu}) = E(\bar{X}) = E\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n} \sum_{i=1}^n E(X_i)$ $= \frac{1}{n} \sum_{i=1}^n \mu = \frac{1}{n} \times n\mu = \mu$	μ = population mean. n = sample size
		Variance of the mean estimator, $\hat{\mu}$	$Var(\hat{\mu}) = Var\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} \left[\sum_{i=1}^n Var(X_i) \right]$ $= \frac{1}{n^2} \left[\sum_{i=1}^n \sigma^2 \right] = \frac{1}{n^2} \times n\sigma^2 = \frac{\sigma^2}{n}$	
		Sample estimator of variance, $\hat{\sigma}^2$	$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \hat{\mu})^2$	
		The unbiased estimator of variance	$s^2 = \frac{n}{n-1} \sigma^2 = \frac{n}{n-1} \times \frac{1}{n} \sum_{i=1}^n (X_i - \hat{\mu})^2$ $= \frac{1}{n-1} \sum_{i=1}^n (X_i - \hat{\mu})^2$	σ^2 = sample variance s^2 = unbiased estimator of variance n = sample size
		The central limit theorem (CLT)	$\frac{\hat{\mu} - \mu}{\frac{\sigma}{\sqrt{n}}} \rightarrow N(0,1)$	μ = population mean. $\hat{\mu}$ = population mean estimator σ = the population standard deviation

	Hypothesis Testing	The test statistic	Test statistic = $\frac{(\text{Sample statistic} - \text{Hypothesized value})}{\text{Standard error of the sample statistic}}$	
		Confidence Interval with two-sided test alternatives	$\left[\hat{\mu} - C_{\alpha} \times \frac{\hat{\sigma}}{\sqrt{n}}, \hat{\mu} + C_{\alpha} \times \frac{\hat{\sigma}}{\sqrt{n}} \right]$	C_{α} = critical value at α test size. n = sample size $\hat{\mu}$ = sample mean
		The confidence interval for one-sided alternative	Lower alternative: $\left(-\infty, \hat{\mu} - C_{\alpha} \times \frac{\hat{\sigma}}{\sqrt{n}} \right)$	C_{α} = critical value at α test size. n = sample size $\hat{\mu}$ = sample mean
			Upper alternative: $\left(\hat{\mu} + C_{\alpha} \times \frac{\hat{\sigma}}{\sqrt{n}}, \infty \right)$	
	Linear Regression	The linear regression model	$Y = \beta_0 + \beta X + \epsilon$	β_0 = the constant intercept β = the slope ϵ = an error
		The ordinary least squares (OLS)	The residual sum of squares is: $\sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta} x_i)^2 = \sum_{i=1}^n \hat{\epsilon}_i^2$	$\hat{\beta}_0 = \bar{Y} - \hat{\beta} \bar{X}$ $\hat{\beta} = \frac{\sum_{i=1}^n (x_i - \bar{X})(y_i - \bar{Y})}{\sum_{i=1}^n (x_i - \bar{X})^2}$

				$\hat{\beta} = \frac{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{X})^2 (y_i - \bar{Y})^2}{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{X})^2} = \frac{Cov(X, Y)}{Var(X)}$ $= \frac{\hat{\sigma}_{XY}}{\hat{\sigma}_X^2} = \frac{\hat{\rho}_{XY} \hat{\sigma}_X \hat{\sigma}_Y}{\hat{\sigma}_X^2} = \frac{\hat{\rho}_{XY} \hat{\sigma}_Y}{\hat{\sigma}_X}$
Regression with Multiple Explanatory Variables	Multiple regression model	$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_k X_{ki} + \epsilon_i, \forall i = 1, 2, \dots, n$		β_i 's = coefficients of the the regression model
	Measures of goodness of fit		$R^2 = r^2$	R^2 = coefficient of variation r = correlation coefficient between X and Y $r = \frac{Cov(X, Y)}{\sigma_X \sigma_Y}$
	Method for regression with one or more than one independent variables	$\sum_{i=1}^n (Y_i - \bar{Y})^2 = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 + \sum_{i=1}^n (Y_i - \hat{Y})^2$		$\sum_{i=1}^n (Y_i - \hat{Y})^2 =$ explained sum of squares (ESS) $\sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 =$ residual sum of squares (RSS) $\sum_{i=1}^n (Y_i - \bar{Y})^2 =$ total sum of squares (TSS)
	Coefficient of determination	$R^2 = \frac{\text{Explained variation}}{\text{Total variation}} = \frac{ESS}{TSS} = 1 - \frac{RSS}{TSS}$		RSS = residual sum of squares ESS = explained sum of squares TSS = total sum of squares

	Regression Diagnostics	Adjusted R^2	$\bar{R}^2 = 1 - \frac{\left(\frac{RSS}{n-k-1}\right)}{\left(\frac{TSS}{n-1}\right)} = 1 - \left(\frac{n-1}{n-k-1}\right)(1-R^2)$	RSS = residual sum of squares ESS = explained sum of squares TSS = total sum of squares n = the number of observations k = the number of independent variables.
		F-statistics	$F = \frac{\left(\frac{ESS}{k}\right)}{\left(\frac{SSR}{n-(k+1)}\right)}$	n = number of observations k = number of independent variables.
		Variance inflation factor for the variable X_j	$VIF_j = \frac{1}{1-R_j^2}$	R_j^2 comes from a regression of X_j on the other variables in the model
		Cook's distance	$D_j = \frac{\sum_{i=1}^n (\bar{Y}_i^{-j} - \hat{Y}_i)^2}{ks^2}$	$\bar{Y}_i^{-j}$ = fitted value of $\bar{Y}_i$ when the observed value j is excluded k = number of coefficients in the regression model s^2 = estimated error variance
	Stationary Time Series	Autocovariance function	$\gamma_{t,\tau} = E[(Y_t - E(Y_t))(Y_{t-\tau} - E(Y_{t-\tau}))]$	t = time τ = lag $\gamma_{t,\tau}$ = autocovariance
		Autocorrelation Function (ACF)	$\rho(t) = \frac{Cov(Y_t, Y_{t-\tau})}{\sqrt{V(Y_t)}\sqrt{V(Y_{t-\tau})}} = \frac{\gamma_\tau}{\gamma_0} = \frac{\gamma_\tau}{\gamma_0}$	

		The 1 st order Autoregressive (AR 1) Model	$Y_t = \alpha + \beta Y_{t-1} + \epsilon_t$	β = AR parameter α = the intercept ϵ_t = the shock.
			$E(Y_t) = \mu = \frac{\alpha}{1 - \beta}$	$E(Y_t)$ = expectation of Y_t
			$V(Y_t) = \gamma_0 = \frac{\sigma^2}{1 - \beta^2}$	$V(Y_t)$ = variance of Y_t
			$\gamma_\tau = \beta^{ \tau } \gamma_0$	γ_τ = autocovariance function under the AR1 model
			$\rho(\tau) = \frac{\beta^\tau \gamma_0}{\gamma_0} = \beta^{ \tau }$	$\rho(\tau)$ = autocorrelation function under the AR1 model
		The Lag Operator	$LY_t = Y_{t-1}$	L = lag operator Y_t = time index at time t Y_{t-1} = time index at time $t - 1$
		The p th order autoregressive model (AR(p))	$Y_t = \alpha + \beta_1 Y_{t-1} + \beta_2 Y_{t-2} + \dots + \beta_p Y_{t-p} + \epsilon_t$	Y_t = time index at time t Y_{t-i} = time index at time $t - i$, $i=1, 2, \dots, p$
	Measuring return, volatility, and correlation	The Simple Returns Method	$R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$	R_t = return at time t P_t = price of an asset at time t P_{t-1} = price of an asset at time $t - 1$

		Return over multiple periods	$R_T = \left(\prod_{t=1}^T (1 + R_t) \right) - 1$	R_T = return over multiple periods R_t = return at time t
		Continuously compounded returns method	$r_t = \ln P_t - \ln P_{t-1}$	r_t = continuously compounded return at time t P_t = price of an asset at time t P_{t-1} = price of an asset at time $t - 1$
		Jarque-Bera test	$JB = (T - 1) \left(\frac{\hat{S}^2}{6} + \frac{(\hat{k} - 3)^2}{24} \right)$	$JB \sim \chi^2_2$ T = sample size S = skewness k = kurtosis
Book 3 Financial Markets and Products	Fund Management	Net asset value, NAV	$NAV = \frac{M - L}{N}$	M = market value of assets at the close of the day L = liabilities N = number of outstanding shares
	Using Futures for Hedging	The optimal hedge ratio/minimum variance ratio	$HR = \rho_{SF} \times \frac{\sigma_S}{\sigma_F}$	ρ_{SF} = correlation between spot prices and the futures prices σ_S = standard deviation of the spot price σ_F = standard deviation of the futures price

		The optimal number of futures contracts needed to hedge an exposure	$N = \beta_{portfolio} \times \left[\frac{portfolio\ value}{value\ of\ futures\ contract} \right]$ $= \beta_{portfolio} \times \left[\frac{portfolio\ value}{futures\ price \times contract\ multiplier} \right]$		
		Adjusting a stock portfolio's beta using stock index futures	$Number\ of\ contracts\ required = (\beta^* - \beta) \left(\frac{P}{A} \right)$		β = portfolio beta β^* = target beta after hedging P = portfolio value A = value of the underlying
	Pricing Financial Forwards and Futures	Forward price		$F = S(1 + R)^T$	S = spot price F = forward price R = risk-free interest rate per year compounded annually Q = Yield T = time to maturity I = income $S = \frac{\{F(1 + Q)^T\}}{(1 + R)^T}$
			The known income case	$F = (S - I)(1 + R)^T$	
			The known yield case	$F = S \left\{ \frac{1 + R}{1 + Q} \right\}^T$	
	Valuing Forward Contracts	Value of a long forward contract	Value of a long forward contract = $\frac{F-K}{(1+R)^T}$		$F = S(1 + R)^T$
		Value of a long forward contract; known income case	Value of a long forward contract = $S - I - \left\{ \frac{K}{(1+R)^T} \right\}$		

		Value of a short forward contract	Value of a short forward contract = $\frac{K-F}{(1+R)^T}$	
		Value of a short forward contract; known yield case	Value of a short forward contract = $\left\{ \frac{S}{(1+Q)^T} \right\} - \left\{ \frac{K}{(1+R)^T} \right\}$	
	Commodity Forwards and Futures	Convenience yield, Y	$Y = \left(\frac{S+U}{F} \right)^{\frac{1}{T}} (1+R) - 1$	Y = convenience yield, T = time to maturity F = forward price R = interest rate $F = (S+U) \times \left(\frac{1+R}{1+Y} \right)^T$ U = present value of storage cost
		Cost of carry, C	Cost of carry, $C = \frac{1+R}{1+Q} - 1 \approx R - Q$	F = forward price R = risk-free interest rate per year compounded annually Q = yield
		Future value of an asset	$F = Se^{(R-Q)T}$	S = present value of an asset F = future value of an asset R = risk-free interest rate per year compounded annually Q = yield
		Future value of an asset, in case of storage costs	$F = Se^{(C-Y)T}$	C = cost of carry Y = convenience yield

	Properties of Options	Lower pricing bound for European calls on:	non-dividend paying stocks	$c \geq \max(S_0 - Ke^{-rT}, 0)$		K = strike price S_0 = value of the stock today. S_T = Price of the stock at time T c = value of a European call option C = value of an American call option p = value of a European put option P = value of an American put option
			dividend-paying stocks	$c \geq S_0 - Ke^{-rT} - De^{-rT}$		
		Lower pricing bound for European puts on non-dividend paying stocks	$p \geq \max(Ke^{-rT} - S_0, 0)$			
		Put-call parity	$c + Ke^{-rT} = p + S_0$			
		Lower bounds for American options	Option	Minimum value	Maximum value	
			American call	$C \geq \max(0, S_0 - Ke^{-rT})$	S_0	
	American put		$P \geq \max(0, K - S_0)$	K		
	Properties of Interest Rates	Future value	$FV = A \left(1 + \frac{R}{m}\right)^{m \times n}$			n = time in years m = compounding frequency per year R = interest rate compounded m times per year
		Future value for continuous compounding,	$FV = Ae^{R \times n}$			FV = future value R = continuously compounded rate of interest

		Relationship between forward rates and spot rates	$1 + f_{t,r} = \frac{(1 + y_{t+r})^{t+r}}{(1 + y_t)^t}$	$f_{t,r}$ = forward rate between time t and $t + r$ y_{t+r} = spot rate at time $t+r$ y_t = spot rate at time t
		Macaulay Duration for a zero-coupon bond	$Duration = \sum_{i=1}^n t_i \left[\frac{c_i e^{-y t_i}}{P} \right]$	t_i = time in years until cash flow c_i is received y = continuously compounded yield on a bond P
		Modified duration	$Modified\ duration = D^* = \frac{Macaulay\ duration}{\left(1 + \frac{y}{n}\right)}$	
		Dollar duration	$DD = D^* \times P_0$	P_0 = bond's market price DD = dollar duration
		DV01	$DV01 = DD \times \Delta y = (D^* \times p_0) \times 0.0001$	
		Effective duration	$Effective\ duration = \frac{BV_{-\Delta y} - BV_{+\Delta y}}{2 \times BV_0 \times \Delta y}$	$BV_{-\Delta y}$ = price estimate if yield decreases by a given amount, Δy $BV_{+\Delta y}$ = price estimate if yield increases by a given amount, Δy BV_0 = initially observed bond price Δy = change in yield expressed in decimal form
		Change in price	Change in price = Duration effect + Convexity effect $= [-Duration \times price \times \Delta y] + [1/2 \times Convexity \times price \times (\Delta y)^2]$	
	Corporate Bonds	Zero-coupon bonds	Original issue discount(OID) $= \text{Face value} - \text{Offering price}$	

		The expected return from a bond	Bond expected return = risk free rate + credit spread – expected loss rate	
	Mortgages and Mortgage-backed Securities	Amount of each payment	$PMT = \frac{\text{Principal}}{\left(\frac{1 - (1 + r)^{-n}}{r}\right)}$	PMT = amount of each payment r = monthly interest rate n = total number of months
		Weighted average coupon (WAC)	$WAC = \sum (\text{Percentage value of each mortgage} \times \text{Respective interest rates})$	
		The percentage value of each mortgage	$\text{Percentage value of each mortgage} = \frac{\text{Value of each mortgage}}{\text{Total value of the portfolio}}$	
	Swaps	Forward rate	$R_{forward} = R_2 + (R_2 - R_1) \frac{T_1}{T_2 - T_1}$	R_i = spot rate corresponding with the year T_i $R_{forward}$ = forward rate between T_1 and T_2
		To the pay fixed, receive floating	$V_{swap} = P_{flt} - P_{fix}$	V_{swap} = value of the swap P_{fix} = PV of fixed-leg payment P_{flt} = PV of the floating leg payment.
		To the pay floating, receive fixed	$V_{swap} = P_{fix} - P_{flt}$	
Book 4 Valuation and Risk Models	Measures of Financial Risk	Volatility of the return, R	$\sigma_R = \sqrt{E(R^2) - [E(R)]^2}$	σ_R = Volatility of the return, R

		The variance of the portfolio expected return	$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2\rho w_1 w_2 \sigma_1 \sigma_2$	σ_1 = standard deviation of the first investment σ_2 = standard deviation of the second investment ρ = correlation between investment the first and the second investment
		VaR for normally distributed loss	$\text{VaR} = \mu_p + \sigma_p U$	U = point in the standard normal distribution that has a probability of $X\%$ of being exceeded μ_p = portfolio mean σ_p = portfolio standard deviation
		Expected Shortfall (ES) for normally distributed loss	$ES = \mu_p + \sigma_p \left(\frac{e^{-\frac{U^2}{2}}}{(1-X)\sqrt{2\pi}} \right)$	X = confidence level U = point in the standard normal distribution that has a probability of $X\%$ of being exceeded μ_p = portfolio mean σ_p = portfolio standard deviation
	Calculating and Applying VaR	VaR for linear derivatives	$\text{VaR}_{\text{linear derivative}} = \Delta \times \text{VaR}_{\text{underlying factor}}$	$\Delta = \frac{\text{percentage change in call value}}{\text{percentage change in stock price}}$
		VaR using Historical Simulation	$\text{VaR}(T, X) = \sqrt{T} \times \text{VaR}(1, X)$	$\text{VaR}(T, X)$ = value at risk for a time horizon of T days and confidence level X

		Expected Shortfall using Historical Simulation	$ES(T, X) = \sqrt{T} \times ES(1, X)$	$ES(T, X)$ = expected shortfall for a time horizon of T days and confidence level X
	Measuring and Monitoring Volatility	Volatility of assets returns in a day, n	$\sigma_n = \sqrt{\frac{1}{m-1} \sum_{i=1}^m (r_{n-i} - \bar{r})^2}$	r_i = return on a day i $r_i = \frac{S_i - S_{i-1}}{S_{i-1}}$ S_i = value of an asset on a close of trading day, i S_{i-1} = value of an asset on a close of trading day, i - 1 $\bar{r}$ = average return from the previous m days defined as: $\bar{r} = \frac{1}{m} \sum_{i=1}^m r_{n-i}$
		Simplified formula for current volatility	$\sigma_n = \sqrt{\frac{1}{m} \sum_{i=1}^m r_{n-i}^2}$	σ_n = current volatility r_{n-1}^2 = squared return on day n-1
		Estimating volatility by Exponentially Weighted Moving Average (EWMA model)	$\sigma_n^2 = w_0 r_{n-1}^2 + \lambda \sigma_{n-1}^2$ assets return	σ_n^2 = variance rate on day n λ = a positive constant that is less than one w_0 = weight applied to the most recent return (i.e., the return on day n-1) $w_0 = 1 - \lambda$

				r_{n-1}^2 = squared return on day n-1
		The updated correlation coefficient under EWMA	$Corr(X_n, Y_n) = \frac{Cov_n}{\sigma_{X_n} \sigma_{Y_n}}$	$Corr(X_n, Y_n)$ = updated correlation coefficient Cov_n = covariance for day n $Cov_n = \lambda Cov_{n-1} + (1 - \lambda) X_{n-1} Y_{n-1}$ X_{n-1} = value of X at day n Y_{n-1} = value of Y at day n σ_{X_n} = standard deviation of X at day n σ_{Y_n} = standard deviation of Y at day n
		GARCH 1,1 model	$\sigma_n^2 = w + \alpha r_{n-1}^2 + \beta \sigma_{n-1}^2$	σ_n^2 = variance rate on day n α = weight given to the most recent squared return. β = weight given to the previous variance rate estimate γ = weight given to long-run average variance rate $w = \gamma V_L$ V_L = long-run variance rate $V_L = \frac{w}{\gamma} = \frac{w}{1 - \alpha - \beta}$

				$\alpha + \beta + \gamma = 1$
		The expected variance rate on day t	$\sigma_{n+t}^2 = V_L + (\alpha + \beta)^t (\sigma_n^2 - V_L)$	σ_{n+t}^2 = expected variance rate on day, t σ_n^2 = variance rate on day n V_L = long-run variance
	Measuring Credit Risk	Expected loss	$EL = PD \times LGD \times EAD$	LGD = loss given default (1-Recovery rate) PD = Probability of default EAD = exposure at default
		Portfolio expected loss	$EL_P = \sum EA_i \times PD_i \times LGD_i$	EL_P = expected portfolio loss
		Unexpected loss	$UL = EA \times \sqrt{PD \times \sigma_{LR}^2 + LR^2 \times \sigma_{PD}^2}$	UL = unexpected loss LR = loss rate $\sigma_{PD}^2 = PD \times (1 - PD)$
		Portfolio unexpected loss	$UL_P = EA \times \sqrt{UL_i^2 \times UL_i^2 + 2\rho UL_i UL_j}$	$UL_i = EA \times \sqrt{PD_i \times \sigma_{LR_i}^2 + LR_i^2 \times \sigma_{PD_i}^2}$
		The mean of credit losses	$p_i \times L_i (1 - R_i)$	L_i = amount borrowed in the i th loan p_i = probability of the default for the i th loan R_i = recovery rate in the event of the default by

				the i th loan
		The standard deviation of credit losses	$\sigma_i = \sqrt{(p_i - p_i^2)[L_i(1 - R_i)]^2}$	σ_i = standard deviation of loss on the i th and j th loan $L_i(1 - R_i)$ = loss if the i th loan defaults
		Credit risk portfolio standard deviation	$\sigma_P = \sqrt{n\sigma^2 + n(n-1)\rho\sigma^2}$	σ_P = portfolio standard deviation
		Standard deviation expressed as the percentage of the size of the portfolio	$\alpha = \frac{\sigma_P}{nL} = \frac{\sigma\sqrt{1 + (n-1)\rho}}{L\sqrt{n}}$	n = number of loans L_i = amount borrowed in the i th loan ρ = correlation between losses on the i th and j th loan
		The Gaussian Copula Model-One factor model	$U_i = a_i F + \sqrt{1 - a_i^2} Z_i$	F = common factor for all U_i and Z_i a_i = parameter values that lie between -1 and +1 $F \sim N(0,1)$ $Z_i \sim N(0,1)$
			$\rho_{U_i U_j} = \frac{E(U_i U_j) - E(U_i)E(U_j)}{\sigma_{U_i} \sigma_{U_j}} = E(U_i U_j) = a_i a_j$	$E(U_i) = E(U_j) = 0$ $\sigma_{U_i} = \sigma_{U_j} = 1$ a_i = parameter values that lie between -1 and +1

		The Vasicek Model	A bank defaults if: $U_i \leq N^{-1}(PD)$	N^{-1} = inverse of the cumulative normal distribution PD = probability of default
			$\rho_{U_i U_j} = a_i a_j = a. a = a^2$	$a_i = a$
			Default Rate as a Function of F $= \frac{N(N^{-1}(PD) - aF)}{\sqrt{1 - a^2}}$	
			Capital requirement for banks under the IRB: $(WCDR - PD) \times LGD \times EAD$	$WCDR$ = worst-case default rate LGD = loss given the default EAD = total exposure at default
		Euler's Theorem	as $\Delta X_i \rightarrow 0$ $F = \sum_{i=1}^n Q_i$	$Q_i = X_i \frac{\Delta F_i}{\Delta X_i}$ ΔX_i = small change in X_i ΔF_i = small change F_i
	Operational Risk	Basic indicator approach	Gross income = Interest earned – Interest paid + Noninterest income	
		Standardized Measurement Approach (SMA)	$Loss\ component = 7X + 7Y + 5Z$	X = all losses Y = losses greater than EUR 10 million Z = losses greater than EUR 100 million

		The probability of n losses	$\Pr(n) = \frac{e^{-\lambda} \lambda^n}{n!}$	λ = average losses n = number of losses
		Logarithm of Mean and Variance Loss Severity	$\text{Logarithm of Mean} = \ln\left(\frac{\mu\sigma}{\sqrt{1+w}}\right)$	σ = standard deviation of the loss μ = mean of the loss size $w = \left(\frac{\sigma}{\mu}\right)^2$
			$\text{Variance} = \ln(1+w)$	$w = \left(\frac{\sigma}{\mu}\right)^2$
		Bank's estimated loss	$\begin{aligned} &\text{Estimated loss for bank A} \\ &= \text{Observed loss for bank B} \\ &\times \left(\frac{\text{Bank A revenue}}{\text{Bank B revenue}}\right)^{0.23} \end{aligned}$	
		Power law	$\Pr(v > x) = kx^{-\alpha}$	v = random variable x = random variable greater than v k = scale factor α = a power
	Bond Yields and Return Calculations	The formula for calculating YTM	$P = \frac{C_1}{(1+y)^1} + \frac{C_2}{(1+y)^2} + \frac{C_3}{(1+y)^3} + \dots + \frac{C_N}{(1+y)^N}$	P = price of the bond C_t = annual cash flow in year t N = time to maturity in years y = annual yield (YTM to maturity)

		YTM of a zero-coupon bond	$YTM = \left(\frac{\text{face value}}{\text{current price of the bond}} \right)^{\frac{1}{\text{year to maturity}}} - 1$	
	Applying Duration, Convexity, and DV01	DV01	$DV01 = -\frac{\Delta P}{\Delta r}$	Δr = the size of a parallel shift in the interest rate term structure measured in basis points ΔP = resultant change in the value of the position being considered
		Effective Duration (D)	$D = -\frac{\Delta P/P}{\Delta r} = -\frac{\Delta P}{P\Delta r}$	Δr = size of a parallel shift in the interest rate term structure measured in basis points ΔP = resultant change in the value of the position being considered
		Effective Convexity	$C = \frac{1}{P} \left[\frac{P^+ + P^- - 2P}{(\Delta r)^2} \right]$	P^+ = value of the position when all rates increase by Δr P^- = value of the position corresponding to the decrease of all rates by Δr
	Binomial Trees	One-step binomial	$f = (f_u p + f_d (1 - p))e^{-rt}$	f = value of the option S = current stock price f_u = value of the stock when the stock price moves up by u f_d = value of the stock when the stock price moves down by d r = risk-free rate of interest t = time to maturity p = risk neutral probability

				$p = \frac{e^{rt} - d}{u - d}$ $u = e^{\sigma\sqrt{t}}$ $d = \frac{1}{u}$
		Two-step binomial	$f = (f_u p + f_d (1 - p))e^{-rt}$	$f_u = (f_{uu}p + f_{ud}(1 - p))e^{-rt}$ $f_d = (f_{ud}p + f_{dd}(1 - p))e^{-rt}$ f = value of the option S = current stock price r = risk-free rate of interest t = time to maturity p = risk neutral probability $p = \frac{e^{rt} - d}{u - d}$ $u = e^{\sigma\sqrt{t}}$ $d = \frac{1}{u}$
	Black Schole's Merton Model	Non-dividend paying stock	$C_0 = S_0 e^{-rT} \times N(d_1) - K e^{-rT} \times N(d_2)$ $P_0 = K e^{-rT} \times N(-d_2) - S_0 e^{-rT} \times N(-d_1)$	P_0 = value of a put option C_0 = value of a call option K = strike price S_0 = current stock price r = risk-free rate of interest T = time to maturity

		Dividend-paying stocks at a dividend-yield, q	$C_0 = S_0 e^{-qT} \times N(d_1) - K e^{-rT} \times N(d_2)$ $P_0 = K e^{-rT} \times N(-d_2) - S_0 e^{-qT} \times N(-d_1)$	P_0 = value of a put option C_0 = value of a call option q = dividend yield $d_1 = \frac{\ln\left(\frac{S_0}{K}\right) + \left[r - q + \left(\frac{\sigma^2}{2}\right)\right] T}{\sigma \sqrt{T}}$
		Options on foreign currency	$C_0 = S_0 e^{-r_f T} \times N(d_1) - K e^{-rT} \times N(d_2)$ $P_0 = K e^{-rT} \times N(-d_2) - S_0 e^{-r_f T} \times N(-d_1)$	$d_1 = \frac{\ln\left(\frac{S_0}{K}\right) + \left[r - r_f + \left(\frac{\sigma^2}{2}\right)\right] T}{\sigma \sqrt{T}}$ $d_2 = d_1 - \sigma \sqrt{T}$ r_f = foreign risk-free rate r = domestic risk-free rate
		Options on futures	$C_0 = F_0 e^{-rT} \times N(d_1) - K e^{-rT} \times N(d_2)$ $P_0 = K e^{-rT} \times N(-d_2) - F_0 e^{-rT} \times N(-d_1)$	F_0 = price of the option. P_0 = value of a put option C_0 = value of a call option $d_1 = \frac{\ln\left(\frac{F_0}{K}\right) + \frac{\sigma^2 T}{2}}{\sigma \sqrt{T}}$ $d_2 = d_1 - \sigma \sqrt{T}$
		Warrants: The price of each employee stock option	$P = \left(\frac{N}{N + M}\right) \times \text{Value of the option}$	P = price of each employee stock option N = total number of shares outstanding M = number of new shares (options) contemplated

Option sensitivity Measures		Greek letters for European options: non-dividend paying stocks		$d_1 = \frac{\ln\left(\frac{S_0}{K}\right) + \left[r + \left(\frac{\sigma^2}{2}\right)\right]T}{\sigma\sqrt{T}}$ $d_2 = d_1 - (\sigma\sqrt{T})$ T = time to maturity S_0 = asset price K = strike price r = continuously compounded risk-free rate σ = stock volatility $N(d_i)$ = cumulative distribution function for a standardized normal distribution variable
		Greek letter	Call option	Put option
		Delta	$N(d_1)$	$N(d_1) - 1$
		Vega	$S_0\sqrt{T} N'(d_1)$	$S_0\sqrt{T} N'(d_1)$
		Gamma	$\frac{N'(d_1)}{S_0\sigma\sqrt{T}}$	$\frac{N'(d_1)}{S_0\sigma\sqrt{T}}$
		Theta	$-\frac{S_0N'(d_1)\sigma e^{-qT}}{2\sqrt{T}} - rKN(d_2)e^{-rT}$	$-\frac{S_0N'(d_1)\sigma e^{-qT}}{2\sqrt{T}} + rKN(-d_2)e^{-rT}$
		Rho	$KT e^{-rT} N(d_2)$	$-KT e^{-rT} N(-d_2)$
		Greek letters for European options: dividend-paying stocks		$d_1 = \frac{\ln\left(\frac{S_0}{K}\right) + \left[r + \left(\frac{\sigma^2}{2}\right)\right]T}{\sigma\sqrt{T}}$ $d_2 = d_1 - (\sigma\sqrt{T})$ T = time to maturity S_0 = asset price K = strike price r = continuously compounded risk-free rate σ = stock volatility $N(d_i)$ = cumulative distribution function for a standardized normal distribution variable $N'(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$ q = dividend yield
		Greek letter	Call option	Put option
		Delta	$e^{-qT} N(d_1)$	$e^{-qT} [N(d_1) - 1]$
		Vega	$S_0\sqrt{T} N'(d_1)e^{-qT}$	$S_0\sqrt{T} N'(d_1)e^{-qT}$
		Gamma	$\frac{N'(d_1)e^{-qT}}{S_0\sigma\sqrt{T}}$	$\frac{N'(d_1)e^{-qT}}{S_0\sigma\sqrt{T}}$
		Theta	$-\frac{S_0N'(d_1)\sigma e^{-qT}}{2\sqrt{T}} + qS_0N(d_1)e^{-qT} - rKN(d_2)e^{-rT}$	$-\frac{S_0N'(d_1)\sigma e^{-qT}}{2\sqrt{T}} - qS_0N(-d_1)e^{-qT} + rKN(-d_2)e^{-rT}$
		Rho	$KT e^{-rT} N(d_2)$	$-KT e^{-rT} N(-d_2)$